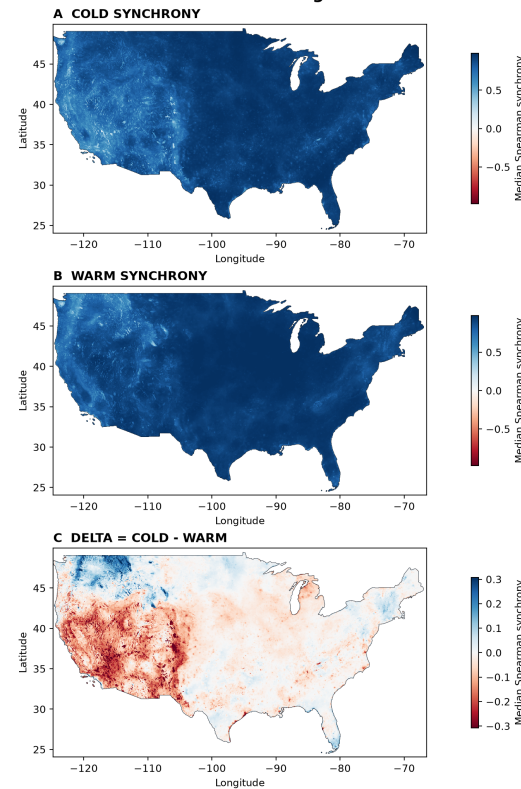


How we built and checked the CONUS synchrony map

A step-by-step walkthrough for the CubeDynamics team

CONUS immediate 100 km local-surface reduction - no stacking



Coverage	Method	Primary finding	Station comparison
481,630 PRISM cells	100 km non-stacked reduction	Median Delta = -0.017	1,500 GHCN stations

Delta = cold synchrony - warm synchrony. Blue Delta is positive; red Delta is negative.

The complete workflow

From daily temperature fields to a validated national synchrony map

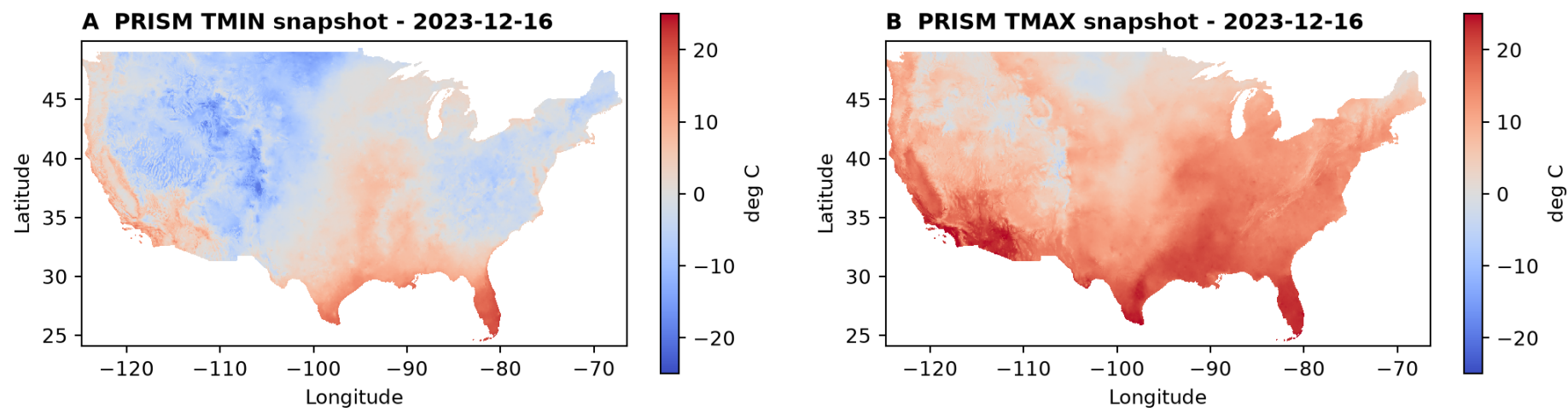


Exact Colorado method preserved: no stacking, no overlap alignment, no second convolution

The national job changes the spatial extent and execution strategy, not the statistic. Every focal pixel is processed independently, its local surface is reduced immediately, and that surface is discarded after the summary is checkpointed.

No stack, overlap alignment, learned model, or second-stage convolution was used.

Step 1 - Freeze the real input data



91 daily labels; frozen 90-day analysis window ending 2024-01-30

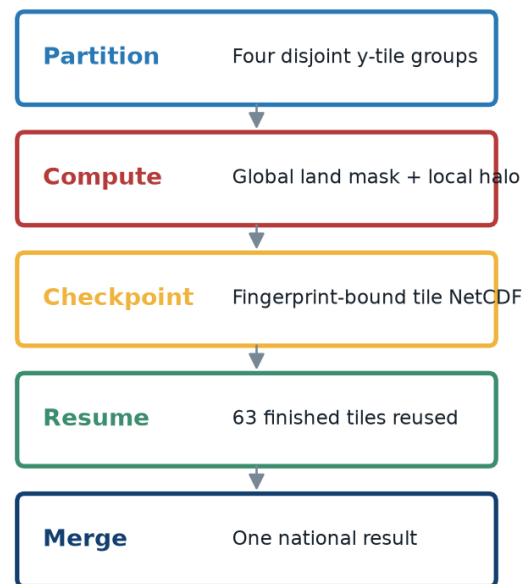
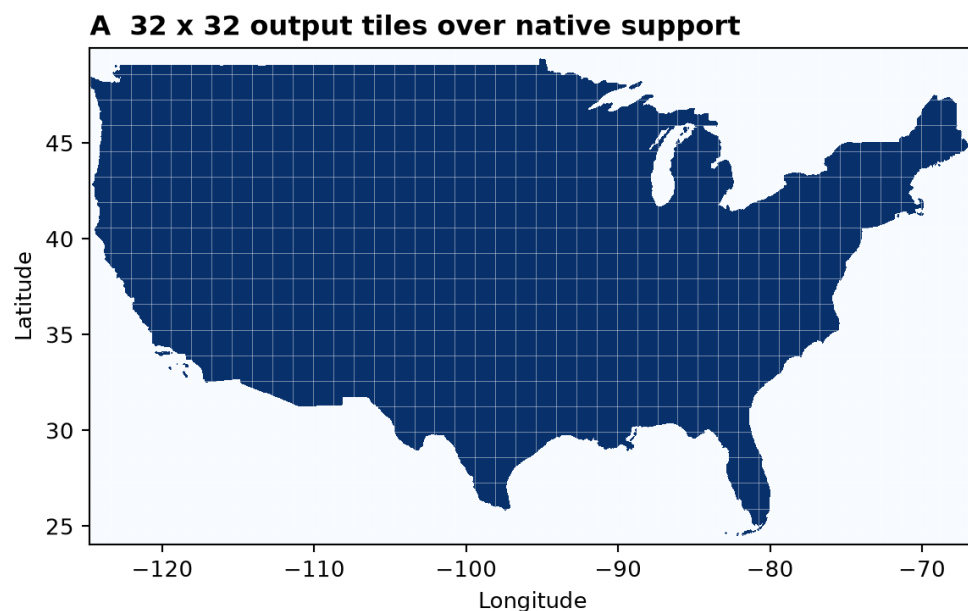


Input contract	Value
Product	PRISM AN daily TMIN and TMAX via NCSCO THREDDs
Date labels	2023-11-01 through 2024-01-30; 91 daily layers
Native grid	621 x 1,399 cells at approximately 1/24 degree
Synthetic fallback	Forbidden for this analysis

Step 2 - Define eligible focal and computation pixels

Scale without changing the science

B Restartable execution path



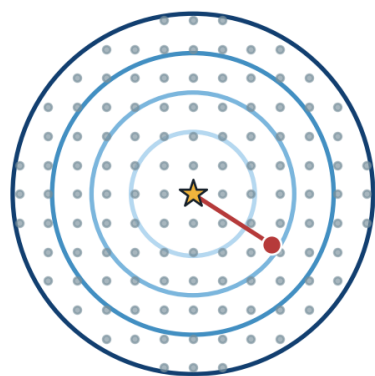
A cell is eligible only when all 91 TMIN and TMAX values are finite. The same global land mask limits pair endpoints; the output mask can then select a partition without changing its neighbors. This separation prevents ocean and nodata cells from affecting tail-support decisions.

At Canada, Mexico, and coast boundaries, the 100 km neighborhood is clipped to native PRISM support. It is not filled with invented observations.

Step 3 - Calculate cold and warm synchrony for each pair

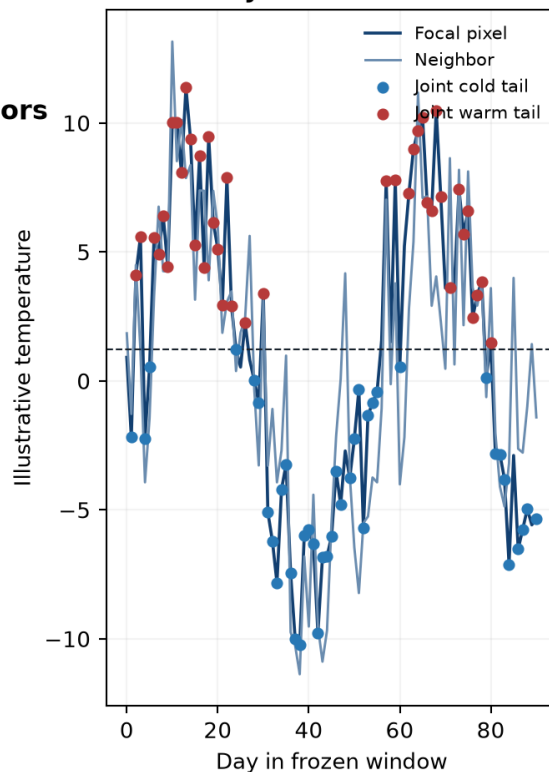
The same pair kernel is applied to every eligible focal-neighbor relationship

A One focal pixel, many neighbors

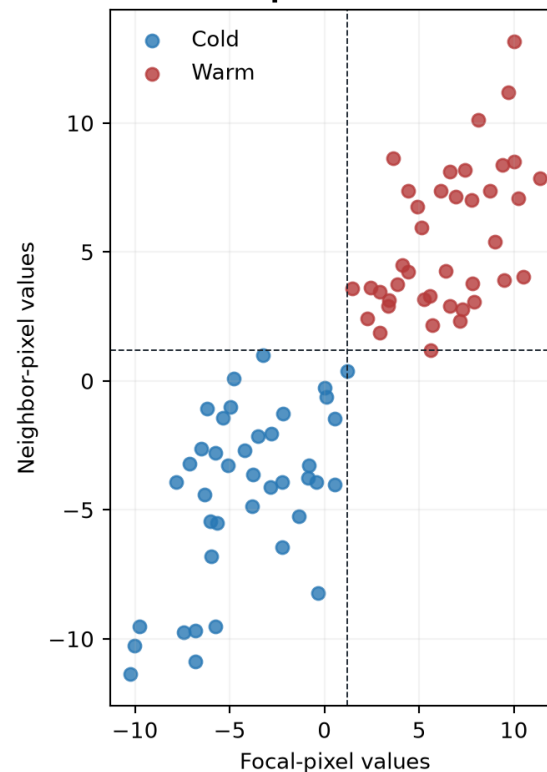


Nested radii: 25 / 50 / 75 / 100 km
The focal self-pair is excluded

B Select joint tails



C Rank dependence within each tail

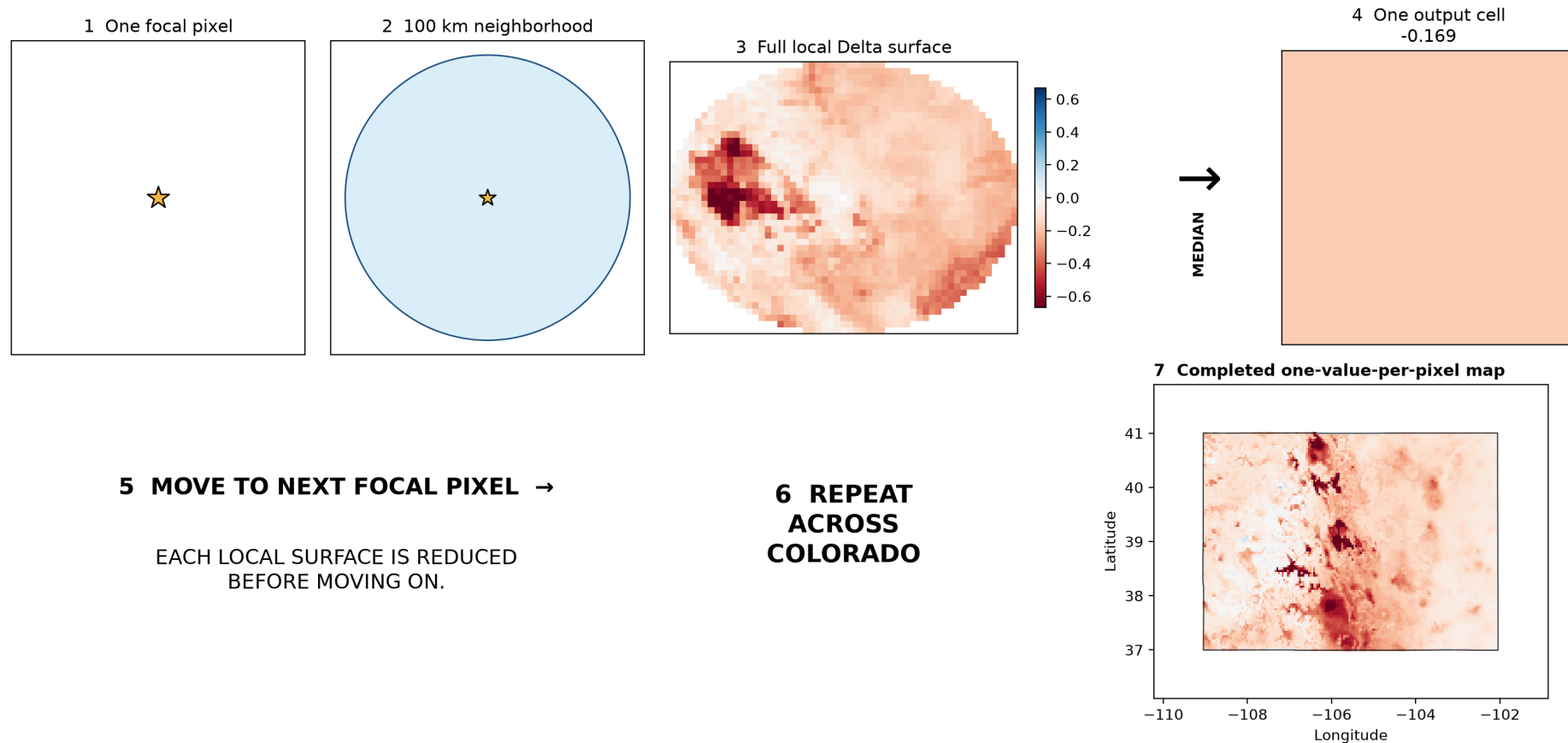


Quantity	Definition
Cold	Spearman dependence on days where both pixels are in their local lower tails
Warm	Spearman dependence on days where both pixels are in their local upper tails
Delta	Cold - warm for the same focal-neighbor pair
Self-pair	Excluded before every reduction

The time-series panel is illustrative; all reported maps use observed PRISM values and the repository's validated pair kernel.

Step 4 - Reduce one local surface to one map cell

NO STACKING

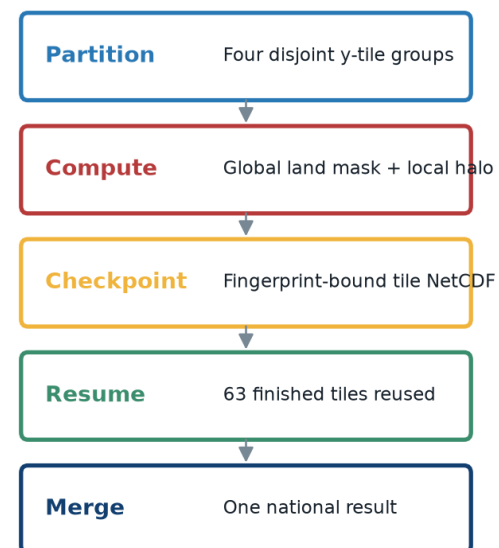
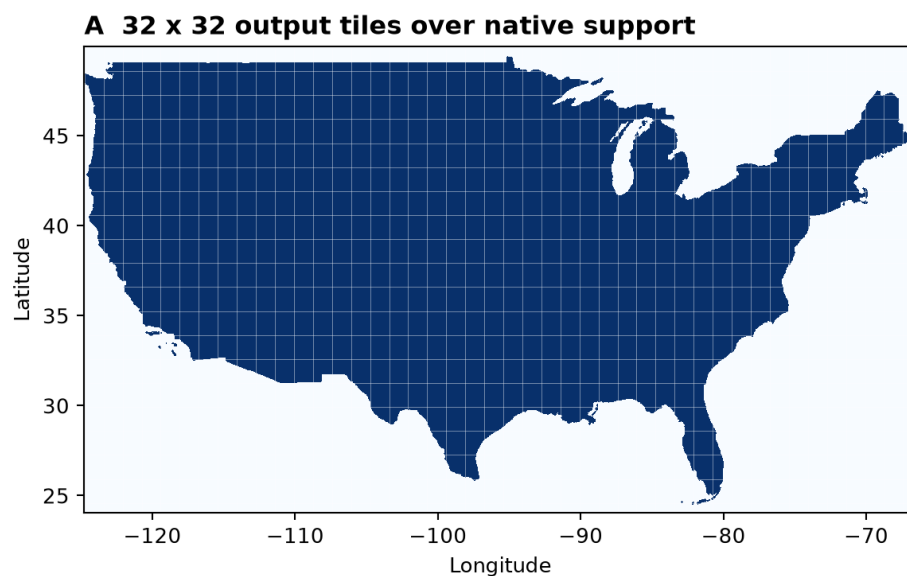


For each focal pixel, all non-self pair values inside the radius form a local surface. The primary map stores median cold, median warm, and median pairwise Delta. Mean, IQR, MAD, extrema, percentiles, counts, and valid fractions are retained as diagnostics. The full surface is not stacked or propagated into a second calculation.

Step 5 - Tile, checkpoint, resume, and merge

Scale without changing the science

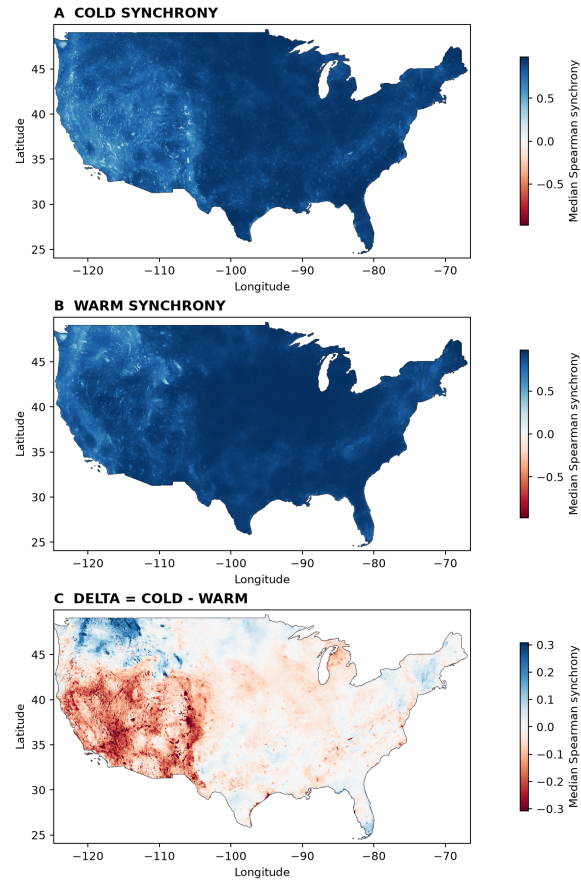
B Restartable execution path



Execution fact	Observed value
Output tiles	555 at 32 x 32 focal cells
Parallel partitions	4 disjoint y-tile groups
Restart behavior	63 tiles reused; 492 computed in the final run
Non-self pair calculations	688,233,788
Slowest partition	27.6 minutes

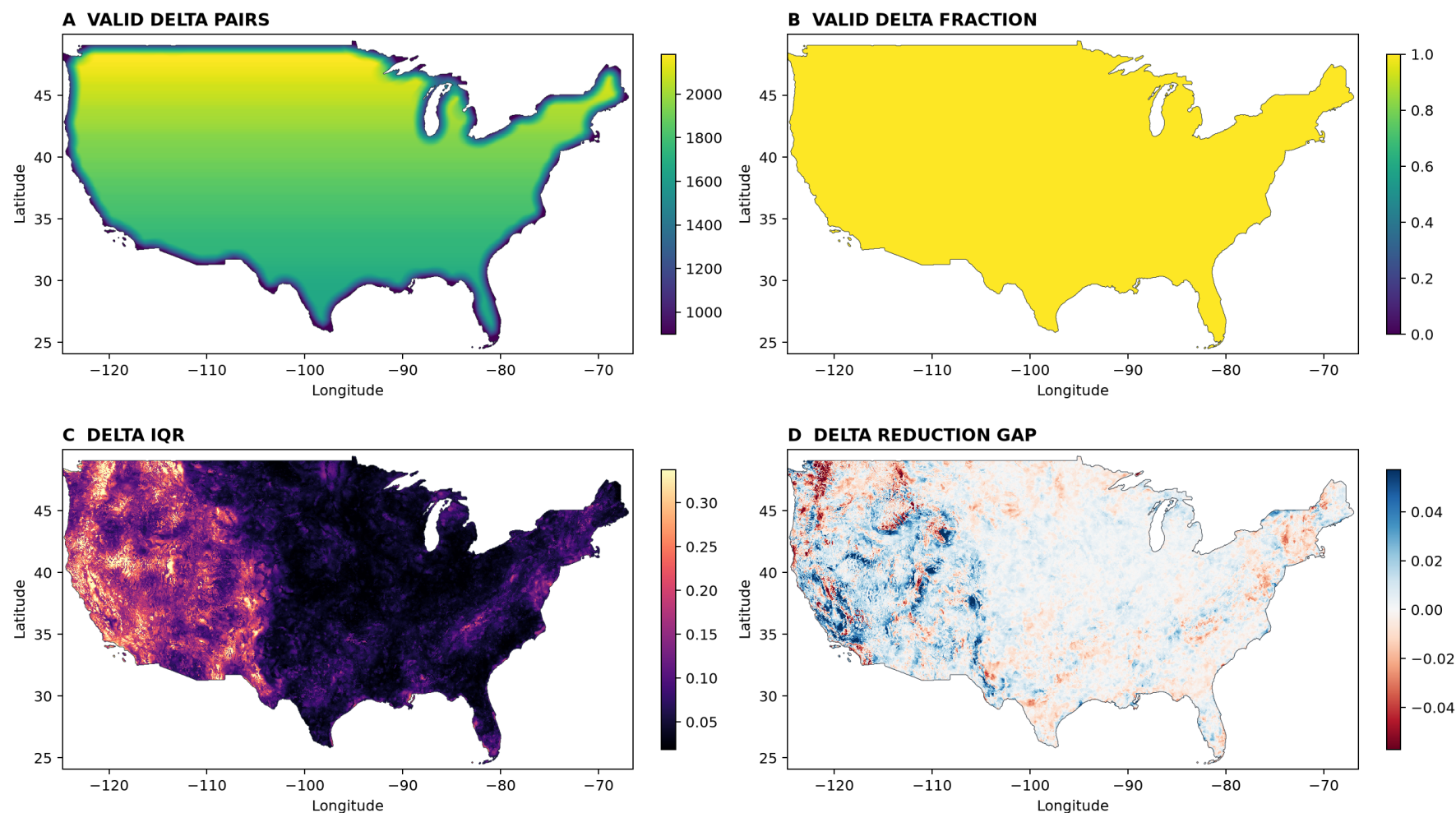
Step 6 - Assemble the national maps

CONUS immediate 100 km local-surface reduction - no stacking



Map	Median across focal pixels	Interpretation
Cold	0.911	Positive cold-tail dependence is widespread
Warm	0.928	Warm synchrony is slightly higher nationally
Delta	-0.017	70.5% of cells are negative

Step 7 - Check support, seams, and reduction behavior

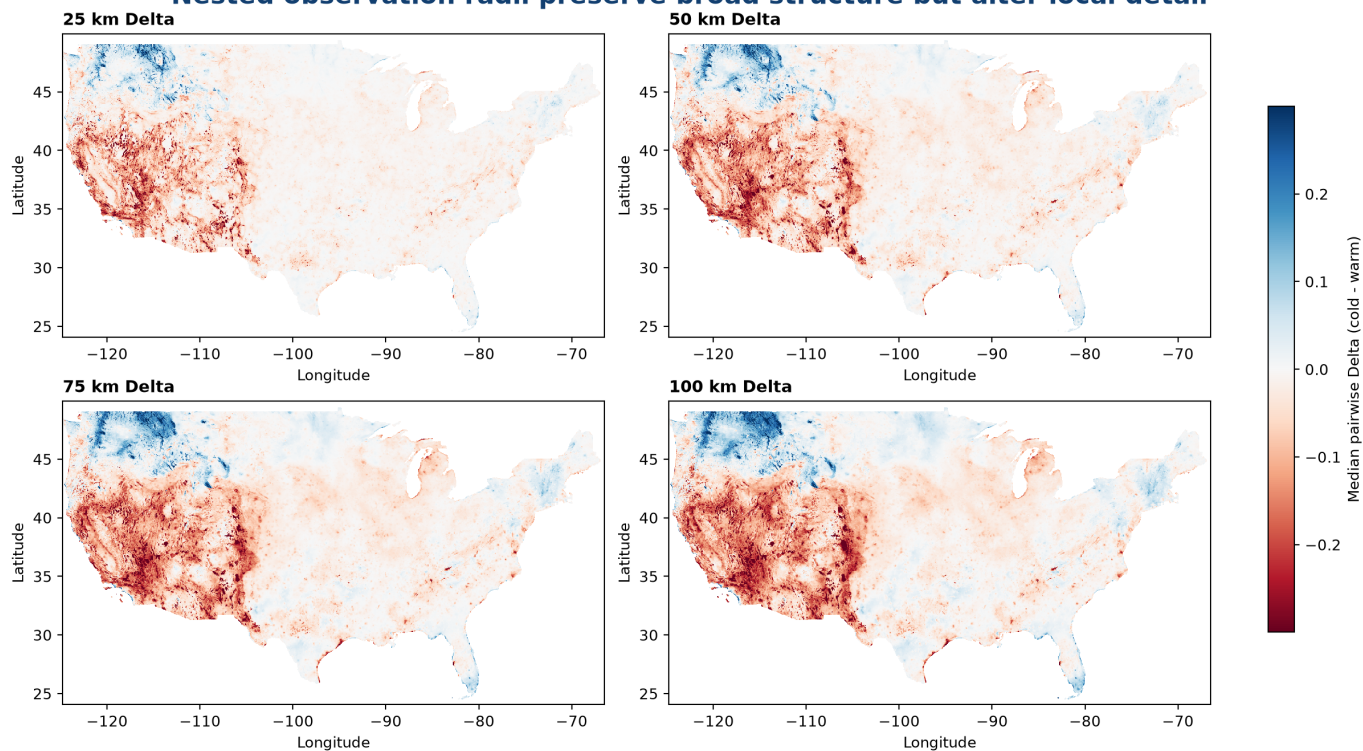


All 481,630 selected pixels have finite primary outputs. Valid pair fractions are 1.0; support counts vary from 19 at clipped boundaries to 2,196 in the interior. Tile-seam jump ratios are approximately one, indicating no numerical checkerboard introduced by checkpoint boundaries.

Pairwise median(cold - warm) and median(cold) - median(warm) are related but not mathematically identical. Their national rank correlation is 0.983; both are retained.

Step 8 - Repeat the reduction at nested radii

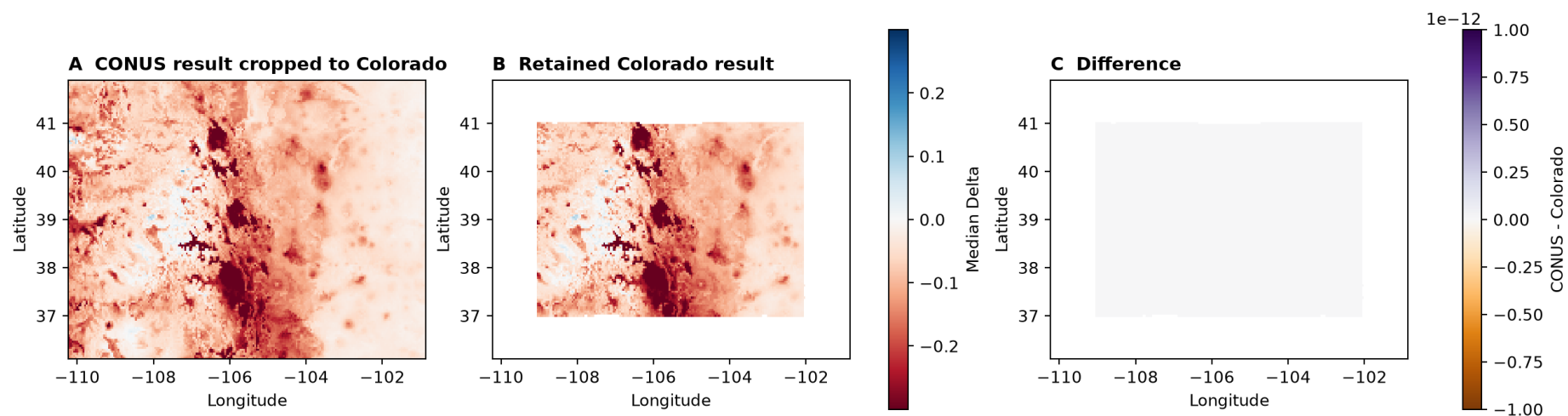
Nested observation radii preserve broad structure but alter local detail



Radius comparison	Delta rank correlation with 100 km
25 vs 100 km	0.798
50 vs 100 km	0.921
75 vs 100 km	0.982

Step 9 - Require exact Colorado equivalence

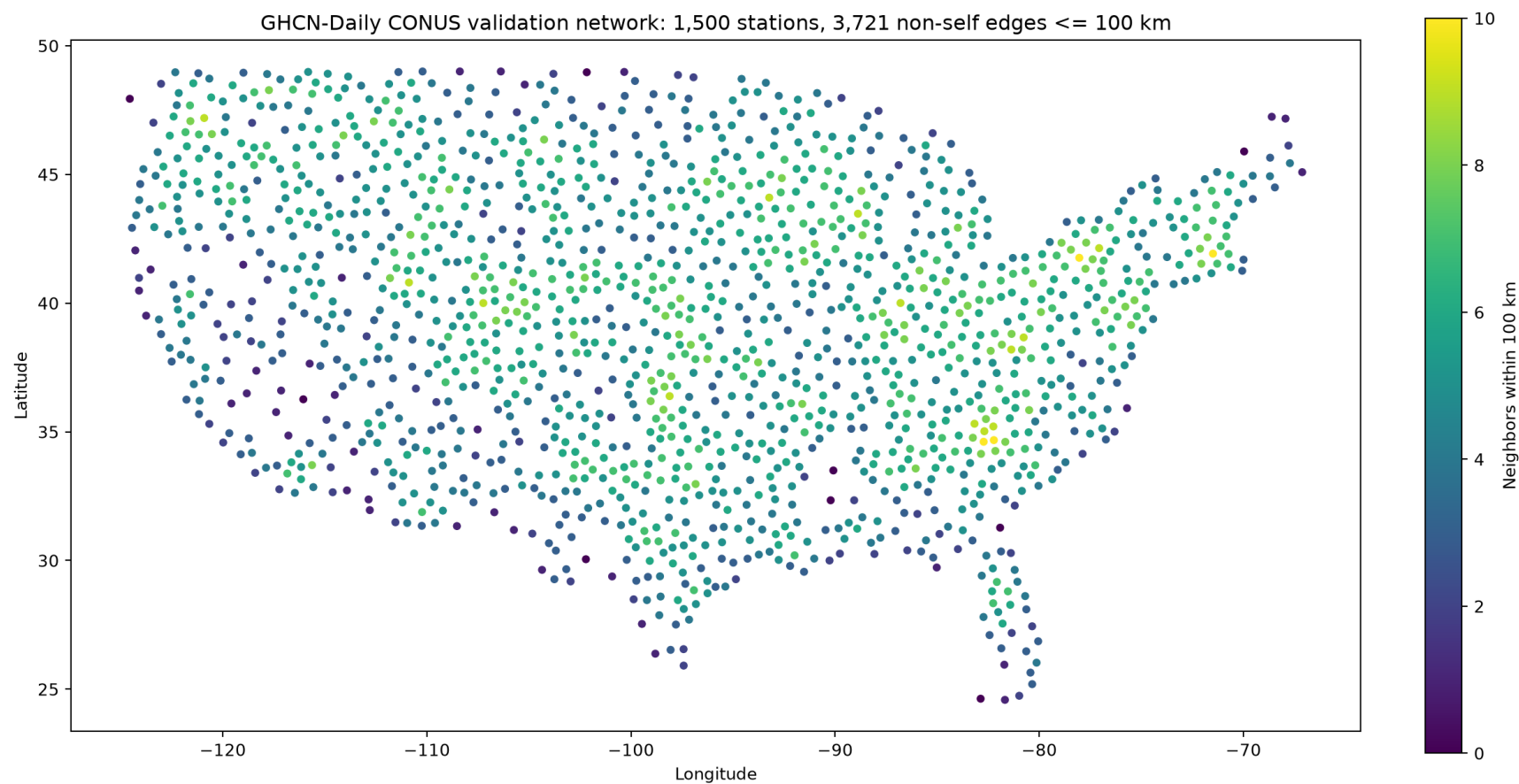
Exact reproduction gate: maximum absolute error = 0.0



The CONUS input exactly matches the retained Colorado-plus-halo snapshot in their common domain. Every nested-radius Colorado output field also matches exactly: maximum absolute error 0.0, with tolerance $1e-12$.

This is the main regression gate showing that scaling to CONUS did not silently change the earlier Colorado method.

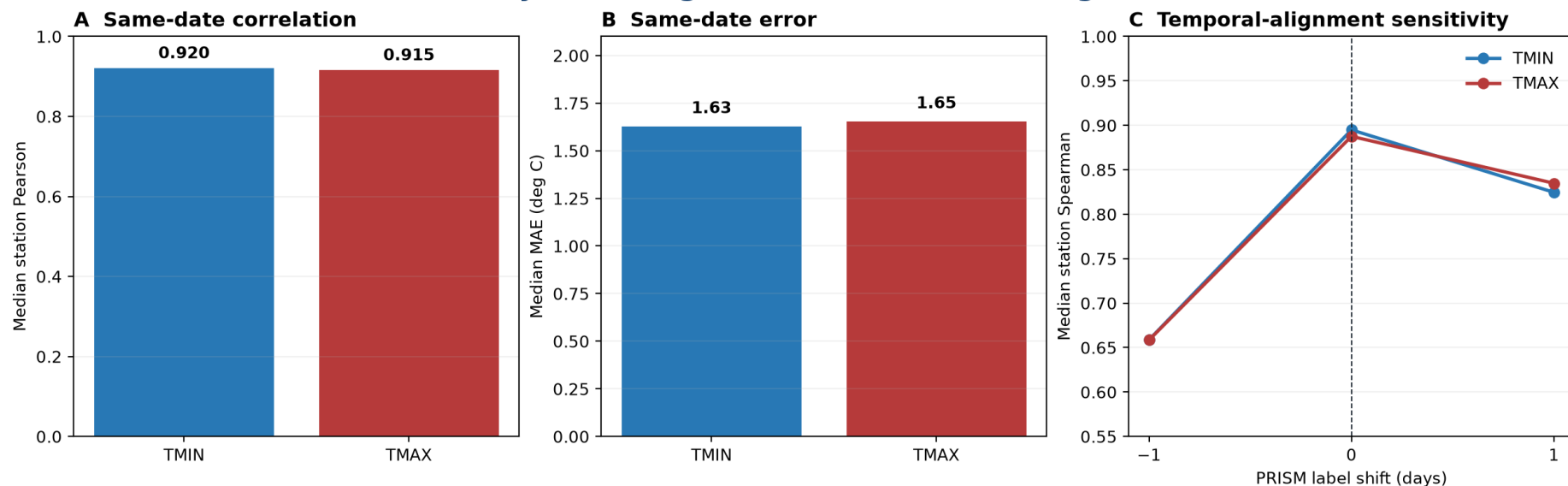
Step 10 - Build a spatially balanced station network



From 1,800 spatially balanced candidates, 1,615 passed the 80% TMIN/TMAX completeness rule and 1,500 were retained. All nonnumeric observations and every nonblank GHCN quality flag were excluded. The network contains 3,721 non-self edges no longer than 100 km.

Step 11 - Check daily station temperatures against PRISM

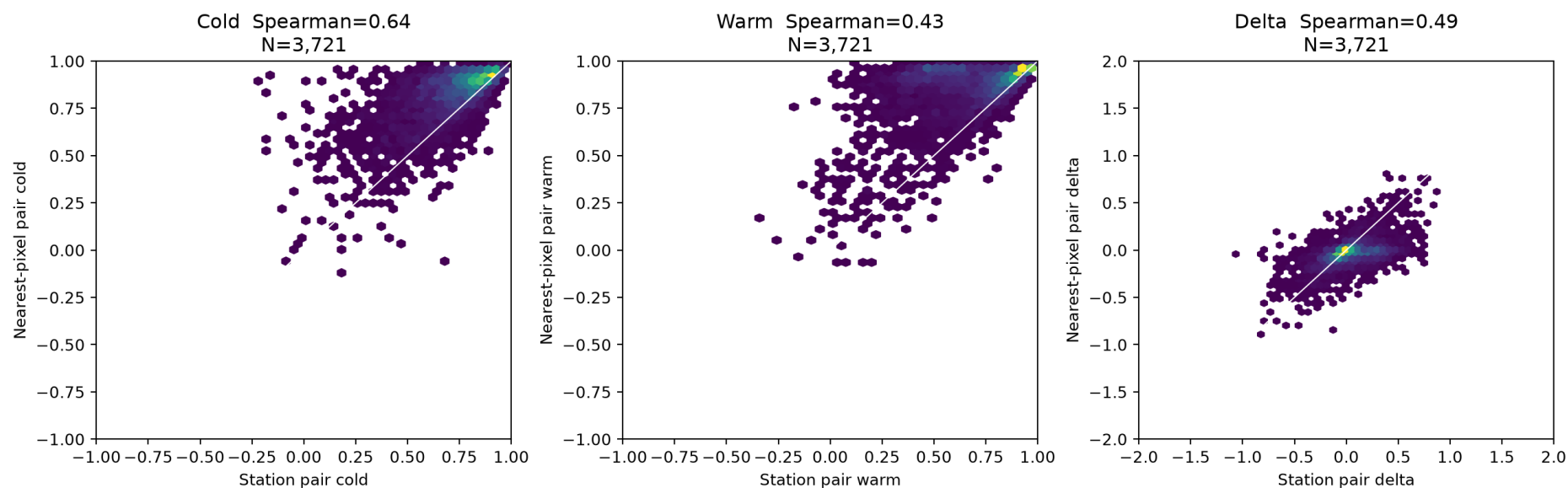
Nearest-cell daily values agree best without shifting the date label



Variable	Median bias	Median MAE	Median Pearson	Median Spearman
TMIN	-0.010 deg C	1.627 deg C	0.920	0.895
TMAX	-0.173 deg C	1.653 deg C	0.915	0.888

The same-date label performs better than shifting PRISM by -1 or +1 day, so no silent time adjustment was introduced.

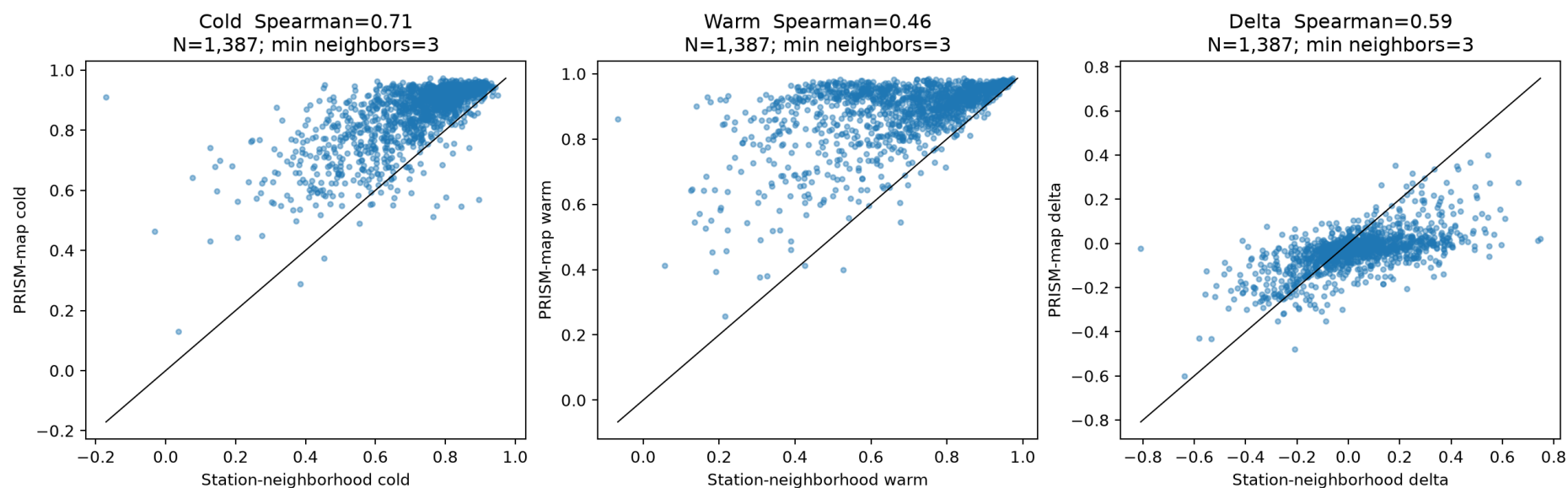
Step 12 - Compare station-pair and nearest-pixel synchrony



Metric	N	Bias	MAE	Pearson	Spearman
Cold	3,721	+0.108	0.124	0.676	0.642
Warm	3,721	+0.179	0.188	0.508	0.429
Delta	3,721	-0.071	0.150	0.543	0.491

These pair rows share stations and are not independent replicates; no edge-level p-values are reported.

Step 13 - Collapse the station network and compare with the map

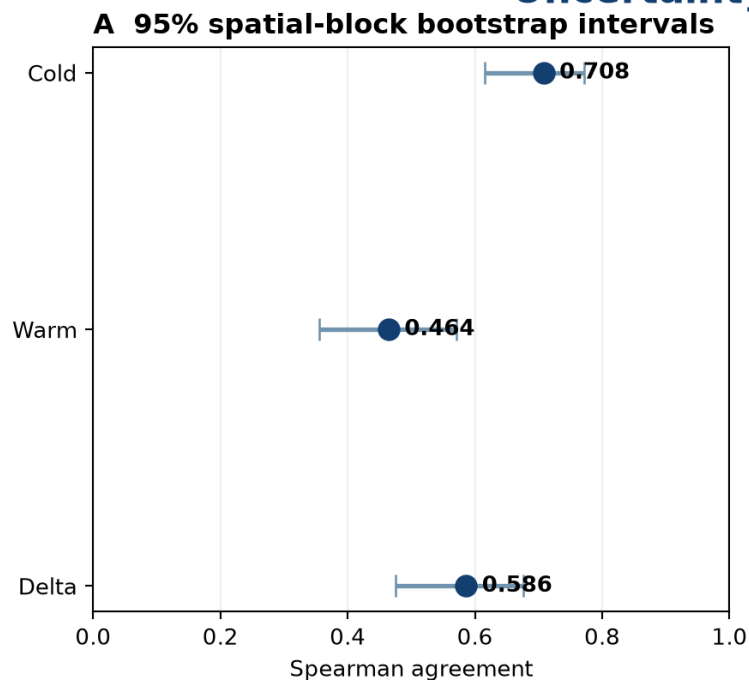


Metric	N stations	Bias	MAE	Pearson	Spearman
Cold	1,387	+0.138	0.142	0.712	0.708
Warm	1,387	+0.196	0.197	0.521	0.464
Delta	1,387	-0.060	0.115	0.620	0.586

The map comparison retains 1,387 stations with at least three finite cold, warm, and Delta incident edges. The station graph is much sparser than the complete raster neighborhood, so perfect one-to-one agreement is not expected.

Step 14 - Quantify uncertainty and state the limits

Uncertainty is reported, not hidden



B How to interpret the evidence

Not an independent holdout

PRISM may ingest some selected stations

Edges share stations

3,721 pair rows are statistically dependent

Different graph density

Stations are sparse and irregular vs. the grid

One winter window

Results are not a climatology or regime map

The 95% intervals resample 5 degree spatial blocks rather than individual stations. This preserves some regional dependence during uncertainty estimation. The correct phrase for these results is *observational agreement*, not *independent validation*.

The maps describe one 90-day winter window. They do not by themselves establish persistent climate regions, causal mechanisms, or a national climatology.

What the team should take away

Completed national calculation

481,630

complete focal pixels

555

restartable tiles

688.2M

non-self pair calculations

4

parallel partitions

27.6 min

slowest partition

15

primary and QC GeoTIFFs

Question	Answer
Did scaling change the Colorado statistic?	No. The exact reproduction gate passed with zero numerical error.
Is the national workflow restartable?	Yes. Every tile is fingerprint-bound, checkpointed, and merge-validated.
Is the national pattern spatially coherent?	Yes. Adjacent-pixel correlations are 0.944 cold, 0.951 warm, and 0.926 Delta.
Do stations agree perfectly?	No. Agreement is strongest for cold, moderate for Delta, and weaker for warm.
What is the strongest caveat?	GHCN is not a strict holdout because PRISM may ingest the same stations.

Products, reproduction, and sources

From daily temperature fields to a validated national synchrony map



Exact Colorado method preserved: no stacking, no overlap alignment, no second convolution

Deliverable	Location or command
National NetCDF	artifacts/nonstacked-conus-baseline/conus_nonstacked_synchrony.nc
GeoTIFFs	artifacts/nonstacked-conus-baseline/rasters/
Station evidence	artifacts/nonstacked-conus-baseline/station_validation/
Reproduce maps	.venv/bin/python scripts/run_nonstacked_conus_baseline.py --stage all
Reproduce station checks	.venv/bin/python scripts/validate_nonstacked_conus_stations.py --stage all
Validation	971 offline tests passed; 5 skipped; 419 deselected

PRISM dataset documentation: https://www.prism.oregonstate.edu/documents/PRISM_datasets.pdf

NOAA/NCEI GHCN-Daily: <https://www.ncei.noaa.gov/products/land-based-station/global-historical-climatology-network-daily>

Analysis generated 2026-09-24 from retained real-data artifacts and machine-readable provenance in the repository.